

Course Title: VLSI System Design

Course Code: EL526

Course Credit: 3-0-2-4

Course Placement: EL526 is the specialization core for the first Semester of M.Tech. ICT students (VLSI and Embedded Systems) and Elective for B.Tech. V (ICT/ICT-CS)

Course Content: The course aims to impart basics of VLSI circuit and system designs. This will be covered by incorporating various practical examples. Different combinational and sequential circuit designs and their co-integration for system implementation will be taught.

Sr. No.	Description
1.	Introduction
	Trends in VLSI/Semiconductor industry, benefits and challenges, ITRS Roadmap, Beyond CMOS, Moore's law, Device scaling effects, Basics of MOSFETs and VLSI, inverter designs, review of MOS analytical models, SPICE models CMOS fabrication process, VLSI circuit design and layout.
2.	Combinational Logic Design
	Static MOS design, Complementary MOS, Ratioed logic, Pass transistor logic, complex logic circuits, transmission gate logic designs.
3.	Sequential Logic Design
	Dynamic logic families and their performance analysis. Static latches, Flip flops and Registers. CMOS Schmitt trigger, Monostable sequential circuits, Astable circuits, CMOS current mirror designs, CMOS amplifiers, Comparator designs, Adder and Multiplier circuits, Memory Designs, Timing analysis: Set-up time (Max-delay) constraint, Hold time (Min-delay) constraint, Clock-skew budgeting, Time borrowing, simple synchronizer, FSM, introduction to pipelined system/ALU, etc.
4.	Datapath Subsystems and Implementation
	Adders, Multipliers, Division, Standard Math function implementation, Comparators, shifter-registers, Error Correcting Codes, FSM, Practical circuit and system designs.
5.	Interconnects
	Interconnect Parameters: Resistance, Inductance, and Capacitance, Interconnect RC delays: Elmore delay calculation. Interconnect models: The lumped RC model, the distributed RC model, the transmission line model. SPICE Wire Models: Distributed RC lines, Transmission line models. Scaling issues in interconnect: Gate and Interconnect delay. Clock distribution system. Crosstalk and noise in interconnects. Crosstalk mitigation techniques. Non-ideal effects in VLSI interconnects and remedial measures. Performance improvement techniques of on-chip interconnects. Advanced interconnect materials and techniques.

Assessment Method: Quizzes, assignments, In-semester examinations and a final end-semester examination will be conducted during the course.

Grading Policy:

In-Sem 1: 20%

In-Sem 2: 20%

End-Sem: 30%

Lab: 10%

Project, Quiz: 20%

Course Outcomes:

After successful completion of this course,

1. Understand concept of MOS transistors, circuit design, their applications in the field of VLSI and semiconductor technology [P1, P2, P3, P4].
2. The students shall be able to design different circuits with optimization with respect to different parameters such as speed, power and area [P3, P4, P12].
3. Use of EDA tools and their efficient execution to implement the system design [P5].

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X							X

Text Books:

1. CMOS Digital Integrated Circuits-Analysis & Design by S.M. Kang & Y. Leblebici, TMH.
2. Principles of CMOS VLSI Design: A System Perspective by NHE Weste & K. Eshraghian, McGraw Hill Pub.
3. Low Power Design Methodologies by Rabaey and Pedram, Kluwer Academic Press.

Reference Books:

1. Microelectronic Circuits by A.S.Sedra and K.C.Smith, Oxford University Press.
2. Integrated Electronics by Millman and Halkias, McGraw Hill Publications.
3. CMOS: Circuit Design, Layout and Simulation by R. Jacob Baker, Harry W. Li, and David E. Boyce, PHI.
4. Introduction to PSPICE by H.M. Rashid, PHI.
5. F.Moll and M.Roca, Interconnection Noise in VLSI Circuits, Kluwer Academic Pub.
6. H.K. Bakoglu, Circuits, Interconnections and Packaging for VLSI, Addison Wesley Pub.